10000 C{}DERS™
Right path for a Bright Career.

DATA SCIENCE / AI

ADDRESS

RD NO 1, KPHB, Kukatpally,
Hyderabad

CONTACT

8790732332, 9700610241

10000 C{}DERS™
Right path for a Bright Career.

About Us

We proudly maintain a **95% placement record**
for students who complete the course, Maintain strong offline attendance, Build Real-World projects, Clear Mock Interviews and clear our CPRP process.
(Communication, Problem-Solving, Resume and Projects)

This all-round preparation ensures they step into the tech industry with confidence, hands-on experience, and job-readiness from day one.

2500+ Students Placed

1600+ Hiring Partners

Recognized for Excellence and Innovation

Highlighting our commitment to excellence and industry leadership by

#startupindia

उद्योग संवर्धन और आंतरिक व्यापार विभाग
DEPARTMENT FOR PROMOTION OF INDUSTRY AND INTERNAL TRADE
सत्यमेव जयते

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Placement Rules and Tips

1. **Attend Regularly –** Be consistent and present for all classes without fail.
2. **Follow Guidance –** Listen carefully to trainers, mentors, and program managers.
3. **Clear L1, L2, and L3 Levels –** Successfully pass these three levels on Communication, problem solving, resume and unique projects to qualify for placement.
4. **Clear Communication Assessment –** Improve your speaking, presentation and interview skills by attending multiple 1-1 reviews with our expert team.
5. **Work on Projects –** Build real-world applications and unique projects using the skills you've learned apart from assignments.
6. **Complete Assignments –** Submit all tasks on time with full effort.
7. **Fast-Track Your Placement –** You don't need to wait 6-7 months for a job! If you clear all reviews and mock interviews, we will send your profile to the placement team immediately.
8. **Practice with Mock Interviews –** Take mock interviews to boost your confidence and performance.
9. **Deploy Your Work –** Upload your projects to GitHub to showcase your expertise.
10. **Prepare a Strong Resume –** Create an ATS-friendly resume and update it based on job requirements.
11. **Job links –** Apply job links regularly.
12. **Stay Active on LinkedIn –** Update your profile and share your work at least three times a week. Be active.
13. **Engage in Learning –** Actively participate in discussions, activities, and seminars.
14. **Pass the Aptitude Test –** Prepare well and clear the required assessment.
15. **Be Available for Opportunities –** Stay responsive and actively engaged until you get placed.
16. **At the time of Placement –** Should attend offline and should be ready to accept and work on the changes/suggestions from the placement team with patience.

Table of CONTENT

01. FOUNDATION & DATA LITERACY

Module 1: Data Science Landscape & Career Paths

- What is Data Science? Real-world impact and applications
- **Career Paths:** Data Scientist vs Data Analyst vs ML Engineer vs Data Engineer
- **Industry Applications:** Healthcare, Finance, E-commerce, Sports, Social Media
- **Success Stories:** Case studies of data-driven decisions
- **Setting Expectations:** Salary ranges, skill requirements, jobmarket.

Module 2: Data Fundamentals & Ethics

- **Types of Data:** Structured vs Unstructured, Quantitative vs Qualitative
- **Data Collection Methods:** Surveys, APIs, Web scraping, Sensors
- **Data Quality:** Accuracy, Completeness, Consistency, Timeliness
- **Data Ethics & Privacy:** GDPR basics, Algorithmic bias, Responsible AI
- **Data Lifecycle:** Collection → Storage → Processing → Analysis → Insights

Module 3: Excel/Google Sheets Mastery

- **Core Functions:** VLOOKUP, INDEX-MATCH, Conditional formatting
- **Pivot Tables & Charts:** Dynamic reporting and visualization
- **Data Cleaning:** Remove duplicates, handle errors, text-to-columns
- **Statistical Functions:** AVERAGE, MEDIAN, STDEV, CORREL
- **Project:** Create a sales performance dashboard

02. PYTHON PROGRAMMING & DATA TYPES

Module 4: Python Programming Fundamentals

- **Environment Setup:** Anaconda, Jupyter Notebooks, VS Code
- **Python Basics:** Variables, data types, operators, input/output
- **Control Structures:** Conditions, loops, nested logic
- **Functions:** Definition, parameters, return values, scope
- **Data Structures:** Lists, tuples, dictionaries, sets
- **File Handling:** Reading/writing CSV, TXT, JSON files
- **Error Handling:** Try-except blocks, debugging techniques
- **Project:** Build a student grade management system

Module 5: Data Manipulation with Pandas

- Series & DataFrames: Creation, indexing, slicing
- Data Loading: CSV, Excel, JSON, SQL databases
- Data Exploration: head(), info(), describe(), shape
- Data Cleaning: Handle missing values, duplicates, data types
- Data Transformation: Filtering, sorting, groupby, pivot tables
- String Operations: Text cleaning, regex basics
- Date/Time Handling: Parsing dates, time series basics
- Project: Clean and analyze a messy real-world dataset

Module 6: Numerical Computing with NumPy

- Array Operations: Creation, indexing, slicing, reshaping
- Mathematical Functions: Statistics, linear algebra basics
- Broadcasting: Efficient array operations
- Integration with Pandas: When to use NumPy vs Pandas
- Project: Financial portfolio analysis with stock data

03. DATA ANALYSIS & VISUALIZATION

Module 7: Exploratory Data Analysis (EDA)

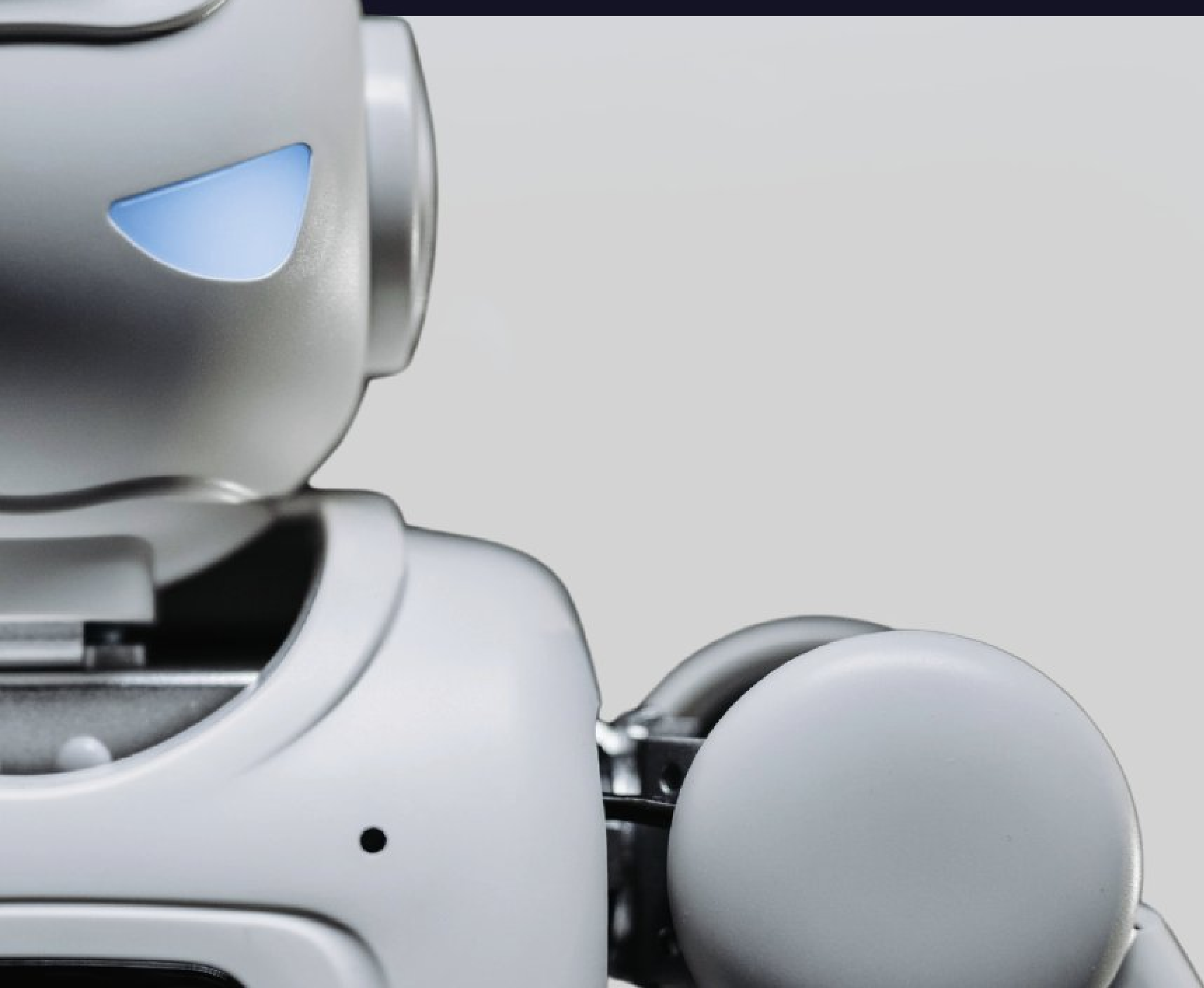
- Statistical Measures: Mean, median, mode, variance, standard deviation
- Distribution Analysis: Histograms, box plots, skewness, kurtosis
- Correlation Analysis: Pearson, Spearman correlation
- Outlier Detection: IQR method, Z-score, visualization techniques
- Data Profiling: Automated EDA tools (pandas-profiling)
- Project: Complete EDA on Titanic/House prices dataset

Module 8: Data Visualization Mastery

- Matplotlib Fundamentals: Plots, subplots, customization
- Seaborn for Statistical Plots: Heatmaps, pair plots, regression plots
- Plotly for Interactive Visualizations: Dynamic charts, dashboards Chart Selection: When to use bar, line, scatter, histogram, etc.
- Storytelling with Data: Design principles, color theory, annotations
- Advanced Visualizations: Geographic plots,
- network graphs Project: Create an interactive dashboard for COVID-19 data

Module 9: Statistics for Data Science

- Descriptive Statistics: Measures of central tendency and spread
- Probability Fundamentals: Basic probability, conditional probability
- Probability Distributions: Normal, binomial,
- Poisson Sampling & Sampling Distributions: Central limit theorem
- Confidence Intervals: Interpretation and calculation
- Hypothesis Testing: t-tests, chi-square tests, ANOVA
- A/B Testing: Design, execution, interpretation
- Project: Design and analyze an A/B test for an e-commerce website



04: DATABASE & SQL SKILLS

Managing and querying structured data

Module 10: SQL for Data Analysis

- Database Fundamentals: Tables, relationships, normalization
- Basic Queries: SELECT, WHERE, ORDER BY, LIMIT
- Aggregation: GROUP BY, HAVING, COUNT, SUM, AVG
- Joins: INNER, LEFT, RIGHT, FULL OUTER joins
- Subqueries: Nested queries, correlated subqueries
- Window Functions: ROW_NUMBER, RANK, LAG, LEAD
- Date Functions: Date arithmetic, formatting, extraction
- Python Integration: SQLite, pandas.read_sql()
- Project: Analyze sales data from multiple tables with complex queries

Module 11: Database Design & Advanced SQL

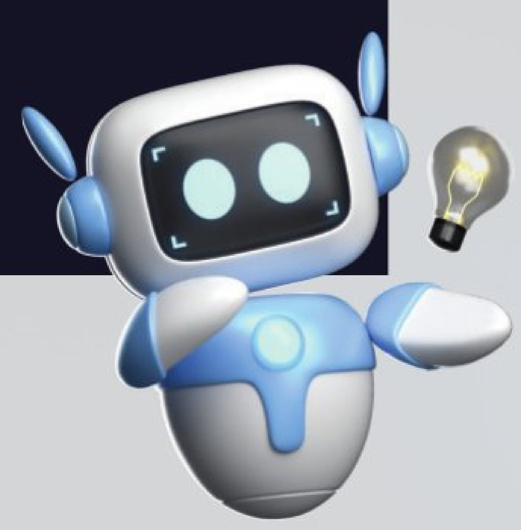
- Data Modeling: ER diagrams, primary/foreign keys
- Performance Optimization: Indexes, query optimization
- Data Warehousing Concepts: OLTP vs OLAP, ETL basics
- NoSQL Basics: When to use MongoDB, document
- databases Project: Design and implement a database for a library management system

🤖 05: MACHINE LEARNING FUNDAMENTALS

Building predictive models

Module 12: ML Foundations & Problem Framing

- What is Machine Learning? Types of learning, real-world applications
- Problem Types: Regression, classification, clustering, recommendation
- ML Workflow: Problem definition → Data → Model → Evaluation → Deployment
- Data Preparation: Feature engineering, scaling, encoding
- Train-Test Split: Validation strategies, cross-validation
- Overfitting vs Underfitting: Bias-variance tradeoff
- Model Evaluation Metrics: Accuracy, precision, recall, F1-score, ROC-AUC



Module 13: Supervised Learning - Regression

- Linear Regression: Simple and multiple regression
- Polynomial Regression: Non-linear relationships
- Regularization: Ridge, Lasso, Elastic Net
- Model Evaluation: R^2, MAE, MSE, RMSE
- Feature Selection: Correlation, statistical tests, recursive elimination
- Project: Predict house prices with comprehensive feature engineering

Module 14: Supervised Learning - Classification

- Logistic Regression: Binary and multiclass classification
- Decision Trees: Splitting criteria, pruning, interpretation
- k-Nearest Neighbors (k-NN): Distance metrics, choosing k
- Naive Bayes: Gaussian, multinomial, Bernoulli
- Support Vector Machines: Linear and non-linear kernels
- Model Evaluation: Confusion matrix, classification
- report Handling Imbalanced Data: SMOTE, class weights, sampling
- techniques Project: Build a customer churn prediction model

Module 15: Ensemble Methods & Model Optimization

- Bagging: Random Forest, Extra Trees
- Boosting: AdaBoost, Gradient Boosting, XGBoost, LightGBM
- Stacking: Meta-learning approaches
- Hyperparameter Tuning: Grid search, random search, Bayesian optimization
- Feature Importance: Tree-based importance, permutation importance
- Model Selection: Comparing multiple algorithms
- Project: Kaggle-style competition with ensemble methods

Module 16: Unsupervised Learning

- Clustering: K-means, hierarchical clustering, DBSCAN
- Dimensionality Reduction: PCA, t-SNE, UMAP
- Association Rules: Market basket analysis, Apriori algorithm
- Anomaly Detection: Isolation Forest, One-class SVM
- Evaluation: Silhouette score, elbow method, cluster interpretation
- Project: Customer segmentation for marketing strategy

06: AI, GENERATIVE AI & LARGE LANGUAGE MODELS

Exploring cutting-edge AI technologies

Module 17: Introduction to Artificial Intelligence & Generative AI

- AI vs ML vs Deep Learning: Understanding the hierarchy
- Evolution of AI: From rule-based systems to neural networks
- What is Generative AI? Text, image, video, and audio generation
- Generative Models: GANs, VAEs, Transformers, Diffusion models
- AI Ethics & Bias: Responsible AI development and deployment
- Current AI Landscape: Major players, models, and applications
- Project: Compare different AI model outputs for creative tasks

Module 18: LLM Fundamentals

- What are LLMs? Architecture and training process
- Transformer Architecture: Attention mechanisms, encoders, decoders
- Pre-training vs Fine-tuning: Understanding the training paradigm
- Popular LLMs: GPT series, BERT, T5, PaLM, Claude, Gemini
- Tokenization: How text is processed by LLMs
- Context Windows: Understanding input limitations
- Project: Build a simple text generation application

Module 19: Working with LLM APIs

- OpenAI API: GPT-3.5, GPT-4, function calling
- Google AI APIs: Gemini, PaLM integration
- Anthropic Claude API: Constitutional AI principles
- Hugging Face Transformers: Open-source model access
- API Best Practices: Rate limiting, error handling, cost optimization
- Prompt Engineering: Crafting effective prompts
- Project: Create a multi-LLM comparison tool

Module 20: Prompt Engineering & LLM Applications

- Prompt Design Principles: Clear instructions, examples, context
- Advanced Techniques: Few-shot learning, chain-of-thought, tree-of-thought
- Prompt Templates: Reusable patterns for different tasks
- LLM Use Cases: Content generation, summarization, translation, coding
- RAG (Retrieval-Augmented Generation): Combining LLMs with knowledge bases
- Function Calling: Integrating LLMs with external tools
- Project: Build a RAG-based question-answering system

Module 21: Vector Databases & Embeddings

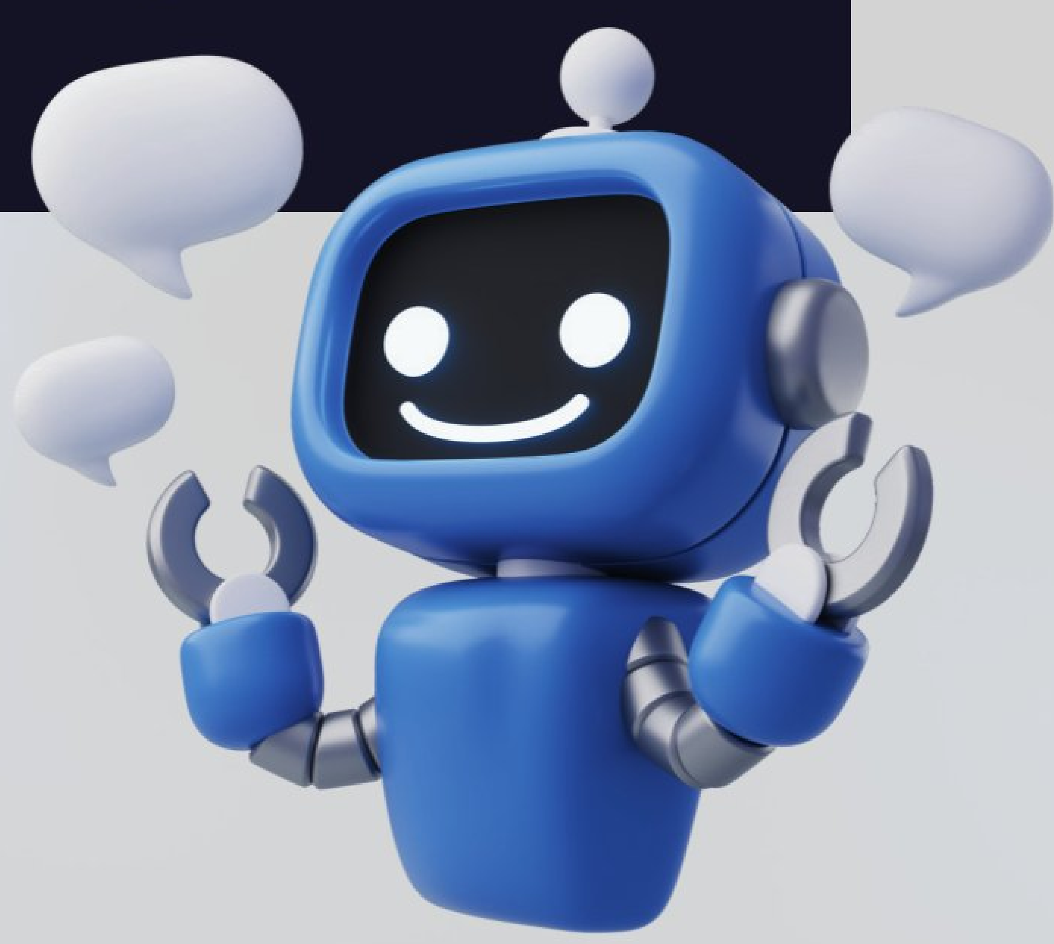
- Text Embeddings: Word2Vec, GloVe, BERT embeddings
- Sentence Transformers: Creating semantic embeddings
- Vector Similarity: Cosine similarity, Euclidean distance
- Vector Databases: Pinecone, Weaviate, Chroma, FAISS
- Embedding Applications: Semantic search, recommendation systems
- Chunking Strategies: Optimal text splitting for embeddings
- Project: Build a semantic search engine for documents

Module 22: Fine-tuning & Model Customization

- When to Fine-tune: Use cases and considerations
- Fine-tuning Approaches: Full fine-tuning vs LoRA vs QLoRA
- Dataset Preparation: Formatting data for fine-tuning
- Training Process: Hyperparameters, evaluation metrics
- Model Evaluation: Benchmarking custom models
- Deployment Considerations: Serving fine-tuned models
- Project: Fine-tune a model for domain-specific tasks

Module 23: Multi-modal AI & Advanced Applications

- Multi-modal Models: CLIP, DALL-E, GPT-4V, Gemini Vision
- Image Generation: Stable Diffusion, Midjourney concepts
- Voice AI: Speech-to-text, text-to-speech integration
- AI Agents: Building autonomous AI systems
- LangChain & LlamaIndex: Frameworks for LLM applications
- AI Safety: Alignment, hallucination detection, content filtering
- Project: Build a multi-modal AI assistant



07: ADVANCED TOPICS & SPECIALIZATIONS

Deep dive into specialized areas

Module 24: Natural Language Processing (NLP)

- Text Preprocessing: Cleaning, tokenization, stemming, lemmatization
- Feature Extraction: Bag of words, TF-IDF, n-grams
- Sentiment Analysis: Rule-based and ML approaches
- Text Classification: Spam detection, topic classification
- Named Entity Recognition: Extracting information from text
- Word Embeddings: Word2Vec, GloVe basics
- Project: Build a movie review sentiment analyzer

Module 25: Time Series Analysis

- Time Series Components: Trend, seasonality, noise
- Stationarity: Tests and transformations
- Moving Averages: Simple and exponential smoothing
- ARIMA Models: Autoregression, differencing, moving averages
- Seasonal Decomposition: STL decomposition
- Forecasting Evaluation: MAE, MAPE, forecast accuracy
- Project: Stock price prediction or sales forecasting

Module 26: Introduction to Deep Learning

- Neural Network Fundamentals: Perceptron, multi-layer networks
- Tensorflow/Keras Basics: Building simple neural networks
- Activation Functions: ReLU, sigmoid, tanh
- Loss Functions & Optimizers: SGD, Adam, learning rate
- Image Classification: CNN basics with MNIST/CIFAR-10
- Transfer Learning: Using pre-trained models
- Project: Build an image classifier for everyday objects

Module 27: Computer Vision Basics

- Image Processing: OpenCV fundamentals
- Feature Detection: Edges, corners, contours
- Image Classification: CNN architectures
- Object Detection: YOLO concepts
- Face Recognition: Basic implementations
- Project: Build a face mask detection system

08: CLOUD & DEPLOYMENT

Making models production-ready

Module 28: Cloud Computing Fundamentals

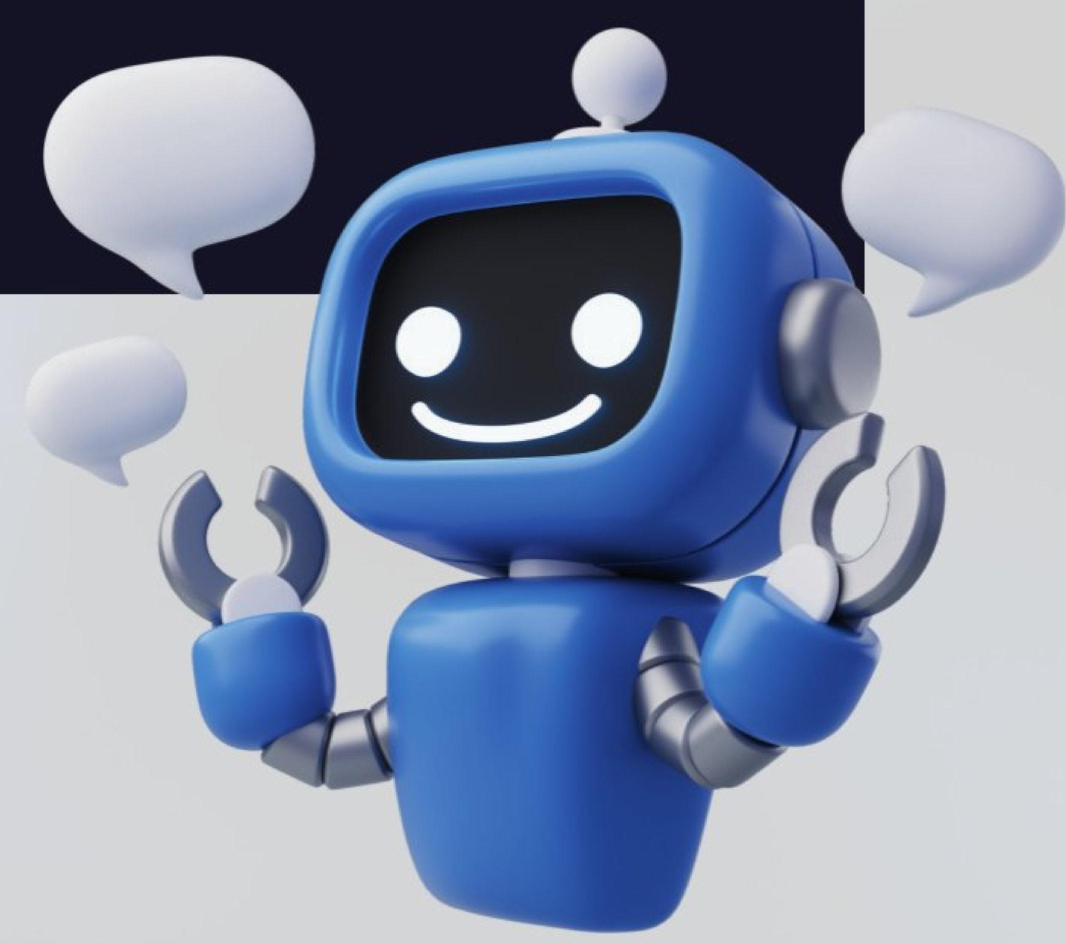
- Cloud Platforms: AWS, Google Cloud, Azure overview
- Cloud Storage: S3, Google Cloud Storage
- Compute Services: EC2, Google Compute Engine
- Managed Services: AWS SageMaker, Google AI Platform
- Cost Management: Understanding pricing, optimization
- Project: Deploy a model on cloud platform

Module 29: Model Deployment & MLOps

- Model Serialization: Pickle, joblib, ONNX
- API Development: Flask, FastAPI for model serving
- Containerization: Docker basics for ML applications
- Model Monitoring: Performance tracking, drift detection
- Version Control: Git for code, DVC for data and models
- CI/CD for ML: Automated testing and deployment
- Project: End-to-end ML pipeline with monitoring

Module 30: Web Applications for Data Science

- Streamlit: Interactive web apps for ML models
- Dash/Plotly: Complex interactive dashboards
- Gradio: Quick ML demos and interfaces
- Heroku Deployment: Free hosting for prototypes
- User Authentication: Basic security concepts
- Project: Deploy a complete ML web application



09: BUSINESS INTELLIGENCE & ADVANCED ANALYTICS

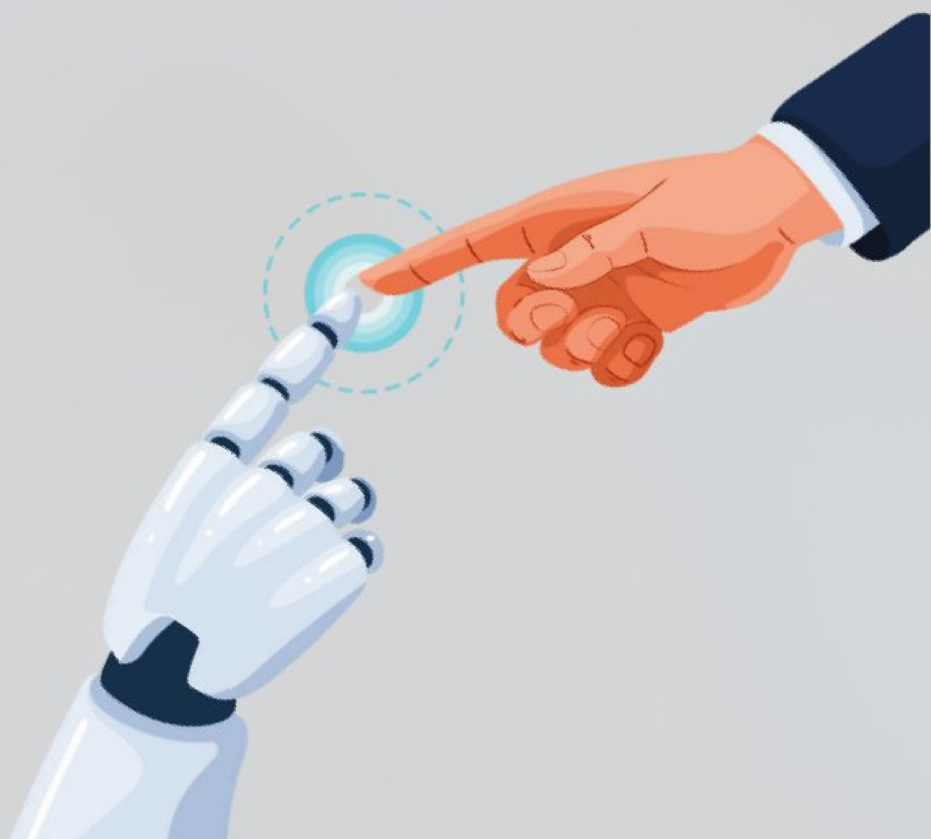
Enterprise-level skills

Module 31: Business Intelligence Tools

- Power BI: Connecting data sources, DAX formulas
- Tableau: Advanced visualizations, calculated fields
- Google Data Studio: Free BI tool mastery
- Dashboard Design: KPIs, drill-downs, interactivity
- Data Storytelling: Executive presentations
- Project: Create comprehensive business dashboard

Module 32: Advanced Analytics & Optimization

- Statistical Modeling: Advanced regression techniques
- Optimization: Linear programming, constraint optimization
- Simulation: Monte Carlo methods
- Causal Inference: Understanding causation vs correlation
- Experimental Design: A/B testing, factorial designs
- Project: Business optimization case study



10: PORTFOLIO & CAREER PREPARATION

Becoming job-ready

Module 33: Portfolio Development

- Profile: Professional setup, README files
- Project Documentation: Clear explanations, code comments
- Jupyter Notebooks: Best practices, storytelling
- Portfolio Website: Showcase projects and skills
- Resume Building: ATS-friendly data science resumes
- LinkedIn Optimization: Professional networking

Module 34: Capstone Project Options

Choose one comprehensive project:

1. E-commerce Recommendation System
2. Healthcare Predictive Analytics
3. Financial Risk Assessment Model
4. Social Media Sentiment Analysis Platform
5. Supply Chain Optimization
6. Real Estate Price Prediction App

Requirements:

- Data collection/cleaning
- Comprehensive EDA
- Multiple ML models
- Model comparison and selection
- Deployment (web app or API)
- Business presentation

Module 35: Interview Preparation & Job Search

- Technical Interview Prep: Coding challenges, ML concepts
- Case Study Practice: Business problem solving
- Behavioral Interviews: STAR method, data science scenarios
- Salary Negotiation: Market research, negotiation tactics
- Job Search Strategy: Where to apply, networking tips
- Mock Interviews: Practice with feedback

🛠 Tools & Technologies Covered

Programming & Development

- Python, SQL, Git, Docker
- Jupyter, VS Code, Google Colab
- Linux command line basics

Data Processing & Analysis

- Pandas, NumPy, Scikit-learn
- SciPy, Statsmodels
- Beautiful Soup, Requests (web scraping)

Visualization & BI

- Matplotlib, Seaborn, Plotly
- Power BI, Tableau, Google Data Studio
- Streamlit, Dash

Cloud & Deployment

- AWS (S3, EC2, SageMaker)
- Google Cloud Platform
- Heroku, Docker

Databases

- SQL (PostgreSQL, MySQL, SQLite)
- MongoDB basics
- Data warehousing concepts

1600+

HIRING PARTNERS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DATA SCIENCE / DATA ANALYTICS (AI & ML)

1 & 2 MONTH

Core Python
SQL
AI Fundamentals
Data Fundamentals
DSA - 1

- COMMUNICATION/ APTITUDE
- WEEKLY TESTS
- REAL TIME PROJECTS

MOCK INTERVIEW - 1

3 & 4 MONTH

ADV Python
EDA + FE
DSA - 2
Data Science Fundamentals
Data Wrangling
Maths (Statistics, Probability)

- COMMUNICATION/ APTITUDE
- WEEKLY TESTS
- REAL TIME PROJECTS

MOCK INTERVIEW - 2

5,6 & 7 MONTH

Machine Learning
AI (llm's + Gen AI)
Deep Learning
Data Science

- COMMUNICATION/ APTITUDE
- WEEKLY TESTS
- REAL TIME PROJECTS

MOCK INTERVIEW - 3

TIMINGS:

8:00 - 10:00 AM
Live Class

10:30 - 11:30 AM
Communication/Aptitude

11:30 - 1:30 PM
Task / Assignmenmt

02:30 - 06:30 PM
Doubts & Practice Session

Timings might change based on Trainers availability

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Everything You Need to Start a Tech Career

SPECIAL OFFER
~~75,000/-~~
42,000/-

- Foundation & Data Literacy
- Python Programming & Tools
- Database & SQL Skills
- Problem Solving + DSA
- Advanced python
- Statistics & Probability (Math)
- Data Analysis & Visualization (EDA, Feature Engineering)
- Business Intelligence & Advanced Analytics
- Machine Learning Fundamentals
- AI & Deeplearning
- LLMs, Gen AI, Prompt Engg
- DataScience
- Portfolio & Career Preparation
- Advanced Topics & Specializations
- Cloud & Deployment (AWS)

Eligible for Roles:

- Data Analyst
- Data Engineer
- Machine learning Engineer
- AI engineer
- Business Analyst
- Database administrator / Architect

Pre Recordings available in Dashboard and having 1 year access after registration.

Contact us :

8790732332, 9700610241, 9381823567

www.10000coders.in

Address

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